Example 12.11

Identify the valid statements from the following sentences.

Solution:

(1) Mount Everest is the highest mountain of the world.

(2) $3+4=8$.

(3) $7+5>10$.

(4) Give me that book.

(5) $(10-x)=7$.

(6) How beautiful this flower is!

(7) Where are you going?

(8) Wish you all success.

(9) This is the beginning of the end.

The truth value of the sentences (1) and (3) are T, while that of (2) is F**.** Hence they are statements. The sentence (5) is true for $x=3$ and false for $x\neq 3$ and hence it may be true or false but not both. So it is also a statement.

The sentences (4), (6), (7), (8) are **not** statements, because (4) is a command, (6) is an exclamatory, (7) is a question while (8) is a sentence expressing one's wishes and (9) is a paradox.

12.3.2 Compound Statements, Logical Connectives, and Truth Tables

Definition 12.8: (Simple and Compound Statements)

Any sentence which cannot be split further into two or more statements is called an **atomic statement** or **a simple statement.** If a statement is the combination of two or more simple statements, then it is called a **compound statement** or a **molecular statement.** Hence it is clear that any statement can be either a simple statement or a compound statement.

Example for simple statements

The sentences (1), (2), (3) given in example 12.11 are simple statements.

Example for Compond statements

Consider the statement, "1 is not a prime number and Ooty is in Kerala".

Note that the above statement is actually a combination of the following two simple statements:

p : 1 is not a prime number.

q : Ooty is in Kerala.

Hence the given statement is not a simple statement. It is a compound statement.

From the above discussions, it follows that any simple statement takes the value either T or F. So it can be treated as a variable and this variable is known as **statement variable** or **propositional variable**. The propositional variables are usually denoted by p, q, r, ...

Definition 12.9 : (Logical Connectives)

To connect two or more simple sentences, we use the words or a group of words such as "and", "or", "if-then", "if and only if", and "not". These connecting words are known as **logical connectives.**

In order to construct a compound statement from simple statements, some connectives are used**.** Some basic logical connectives are **negation (not), conjunction (and)** and **disjunction(or).**

12th_Maths_Vol 2_EM_CH 12_Discrete Mathematics.indd 238 16-12-2022 16:54:32